ERC Consolidator Grant 2027: Research Proposal, Part B1

Acronym: HOVER

Title: Hoverfly-Optimised Vision through Event-driven Recurrent spiking circuits: from a sexually dimorphic eye to active machine vision

Principal Investigator: Prof. Dr. Étienne Serbe-Kamp

Host institution: IU Internationale Hochschule GmbH, Germany *[confirm exact legal name]*

Proposal duration: 60 months

Primary panel: LS5, Neuroscience and Disorders of the Nervous System · **Secondary panel:** PE6, Computer Science and Informatics

Abstract

Seeing is active: eyes move, and how they should move depends on the task. Machine vision is rediscovering this with event cameras, which are blind to static scenes unless the sensor moves. Flies solve the problem on a few milliwatts, and evolution has left a clean experiment. Male hoverflies (Eristalis tenax) chase mates in aerial pursuits; females do not. Males have a specialised dorso-frontal eye zone, a sexually dimorphic motion-sensitive neuron and, as we discovered, a male-only third muscle that moves the retina beneath the lenses. Nobody knows what it is for. HOVER will find out, explain why the male's task demands it, and test whether the answer is a design principle for machine vision. We will (1) build the first connectome-constrained spiking model of a complete gaze loop, from photoreceptors to eye movement, including the electrical synapses we found between motion-sensitive neurons and eye-muscle motor neurons, in the female and the new male Drosophila connectome; (2) map the model onto male and female hoverflies, let it design the stimuli that single out the third muscle, and test them with retinal tracking in closed loop and muscle recordings, while tracing the muscle's developmental and genetic origin; (3) train the models on pursuit versus stabilisation under identical wiring and energy budgets, to test whether the male features emerge as task-optimal solutions; (4) transfer the principles to event cameras that move their sensors, with spiking networks on neuromorphic hardware, benchmarked against deep networks on small-target detection, ego-motion estimation and closed-loop flight; and (5) validate every model on low-cost spiking robots run by students and citizens. HOVER delivers the first photon-to-eye-movement model of a visual circuit, the function and origin of a sex-specific eye muscle, and open, tested design principles for efficient active machine vision.

Why a second panel

HOVER asks a question of systems neuroscience: why is the hoverfly visual system sexually dimorphic, and which circuit motifs make it task-specific (LS5_16, LS5_7)? Two of its five objectives, however, are tested in machines. Objective 4 builds event-based spiking vision systems with task-dependent sensor micro-movements and benchmarks them, on public event-camera datasets and on neuromorphic hardware, against deep-network baselines (PE6_11, PE6_14). Objective 5 runs the models on robots at scale. Reviewers with expertise in computer vision and neuromorphic computing are needed to judge the benchmark design, the baselines and the hardware claims of that part of the work.

HOVER: Hoverfly-Optimised Vision through Event-driven Recurrent spiking circuits

PI: Étienne Serbe-Kamp · **Host:** IU Internationale Hochschule (Germany) · **Partners:** MPI for Biological Intelligence, LMU Munich · **Duration:** 60 months · **Panel:** LS5 (secondary PE6)

1. The question

Seeing is active. Animals move their eyes to stabilise the image, to follow a target and to refresh what they see, and how they move them depends on the task. Machine vision is rediscovering this [112]: event cameras, the neuromorphic sensors that report only changes of brightness, are blind to a static scene unless the sensor moves [53,54]. How should a sensor move for a given task, and which circuit should read it? Flies answer both questions with about 10^5 neurons and a few milliwatts, and evolution has left a clean experiment. Male and female hoverflies share a genome and a brain plan, but only males chase mates in high-speed aerial pursuits, and only males, as we discovered, carry **a third muscle that moves the retina** (Fig. 1). Nobody knows what it is for.

> **HOVER will find out what the third eye muscle is for, explain why the male's task demands it, and test whether the answer is a design principle for machine vision.**

It works in a loop (Fig. 1). A spiking model of the complete fly gaze circuit, built from the connectome, predicts the stimuli that single out the muscle (WP1). Hoverflies see these stimuli while their own eye movements steer the search for better ones (WP2). The model is trained for the male or the female task, to see whether the muscle emerges (WP3). The resulting principles drive event cameras that move their sensor (WP4), and students and citizens test every model on low-cost robots (WP5).

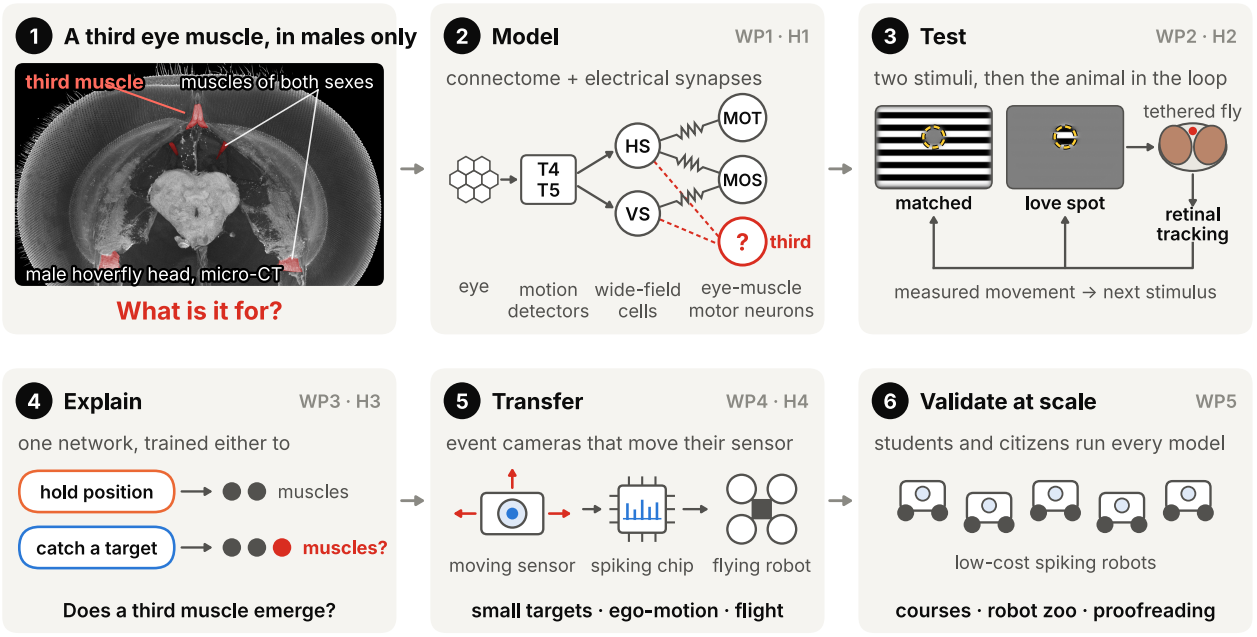


Figure 1. *HOVER in six steps.* (1) Micro-CT of a male hoverfly head, retinal muscles in red; the third muscle, between the two eyes, exists only in males [2]. (2) The gaze circuit as a spiking model built from the connectome, with electrical synapses (zigzags); what drives the third muscle is unknown. (3) Two stimuli on one display, then a stimulus search steered by the tracked retinae. (4) One network trained for the female or the male task. (5) Event cameras that move their sensor, on neuromorphic chips and a quadrotor. (6) Every model runs on low-cost spiking robots.

2. A natural experiment: one genome, two visual machines

Male hoverflies (*Eristalis tenax*) hover on station and dart after passing conspecifics on interception courses [71,72]; females hover, feed and lay eggs. The male visual system is specialised at every level. A dorso-frontal "bright zone" of enlarged facets gathers more light and gives faster, more contrast-sensitive motion detection [15]. It is the hoverfly version of the "love spot" of houseflies and blowflies, where photoreceptor speed is matched to chasing [76] and male-specific neurons exist [74,75]. Hoverflies have four horizontal-system cells, wide-field motion neurons that stabilise gaze in other flies. In males one of them, HSN, has its receptive field narrowed to the bright zone and responds to small moving targets [1]. Descending neurons differ between the sexes at high velocities [23], and hoverfly small-target detectors [16,62] and the male pursuit itself [21] have been measured in the laboratory.

Our own recent work adds the missing level: the effector. Flies move their retinae beneath the stationary lenses with small muscles, which produces an optokinetic reflex and spontaneous, saccade-like shifts [7]. In a comparative survey of 167 insect species by micro-CT and synchrotron tomography, with behaviour in 59 of them, we found retinal movements only in Diptera, reconstructed their stepwise evolution and discovered in hoverflies **a male-only third retinal muscle, the M. interocularis (MIO)**, lodged between the two nearly fused male eyes, next to the bright zones [2]. The male hoverfly therefore has not only a different sensor and a different circuit but a different *active-vision effector*. This is a natural knockout experiment with its rationale built in: the female shows the gaze system without the muscle, the male with it, and the male's extra task is known. HOVER uses it three times: as a test of mechanism (what drives the muscle, and what does it do?), as a test of optimality (does the muscle emerge in a model trained for pursuit?) and as a source of design principles for machines.

3. State of the art and three gaps

Gap 1. Connectomes have outpaced models, and both miss electrical synapses. Synapse-level wiring diagrams now exist for the female *Drosophila* brain [12,47,48], for brain and nerve cord [49] and, since September 2026, for the complete male central nervous system [39]. The models built on them predict neural activity [45] and sensorimotor processing [46], but they (i) use chemical synapses only, because gap junctions are invisible in these electron-microscopy volumes [50], (ii) stop before the motor output and (iii) describe one sex of one species in a passive setting. We have shown why this matters for gaze: the motor neurons that move the *Drosophila* retina receive only 3 % of their chemical synapses from visual neurons, yet they are electrically coupled to the HS and VS cells, and this wiring alone predicts their direction tuning, the two-dimensional optokinetic reflex ($R^2 = 0.99$) and an efference copy that flows back into the visual neurons during spontaneous eye movements [14]. To find electrical synapses at connectome scale I built **MESH**, an open pipeline that detects membrane appositions between neuron meshes, validates them in the electron-microscopy volume and serves them for proofreading [41]: 49 validated contact sites out of 619 candidates for the 26 neurons of the gaze circuit.

Gap 2. Active vision is missing from circuit models and from event-based machine vision. Retinal micro-movements refresh the image, defeat photoreceptor adaptation and sharpen acuity [7,51,52]. Event cameras have the mirror-image problem: they go blind on static scenes [53], although they now reach automotive products [108]. Engineers have begun to add artificial microsaccades with a rotating prism [54], and spiking networks on neuromorphic chips estimate optic flow and fly drones with a few milliwatts [55,56]. But these movement policies are fixed and task-agnostic, and no model links a sensor movement to the circuit that reads it. Our simulations show why this matters (Fig. 2c): the movement that lets an event camera register a static scene hides a small moving target. The hoverfly, which stabilises with two muscles and pursues with three, is nature's worked example of a question engineers have only begun to ask: should a pursuit system move its sensor differently from a stabilising one?

Gap 3. Finding out what a circuit element is for needs stimuli chosen by models, and by the animal. Standard gratings rarely discriminate competing circuit hypotheses; stimuli optimised so that models disagree can [30], and closed-loop methods that let neural responses choose the next image have found what visual neurons respond to best [110,111]. No such method exists for an eye movement. We have built its two parts: a model from photoreceptor to eye muscle that designs the stimuli before an animal is used, and a search in which the tracked retinal movement chooses the next stimulus (section 4).

What machines need from this. Most spiking vision networks are deep networks converted to spikes [64]. They lack the motifs that make fly vision fast, sparse and robust: sensor movements matched to the task [7], an efference copy that tells self-generated from external motion [14], electrical coupling that pools signals without synaptic delay [14,92], push-pull opponent coding [26] and contrast normalisation [38,96]. Whether these transfer can only be decided with a model of a complete biological loop and a benchmark with matched baselines. HOVER provides both.

4. Preliminary results

A model from photoreceptor to eye muscle. We have built the model that WP1–WP3 refine: compound-eye sampling [109], motion detectors with opponency [26], electrically coupled wide-field cells and spiking motor neurons from MESH, and the muscle model of [14]. Fitted to *Drosophila* with two parameters, it reproduces the preferred directions of both retinal motor neurons (212° and 146°; measured 212° and 141°) and the direction of the optokinetic reflex (mean error 17°).

Two stimuli single out the third muscle (Fig. 2a,b). In the hoverfly version, with both eyes, the narrowed HSN drives the third muscle. The model then designs two stimuli for one display. A square-wave

grating that spares the love spot moves the retinae of both sexes alike (within 5 %), although their HSN cells differ: it calibrates each animal and tests the shared part of the circuit. The same grating shown only in the love spot moves the male retina 2.9 times as far as the female's, and 88 % of that difference disappears when the muscle is cut in the model. A small patch shown anywhere else leaves the sexes alike (Fig. 2b). If the muscle were driven by the vertical cells instead, as an amplifier of the stabilising reflex, the pattern would reverse (matched stimulus: male 1.9 times the female; love spot: 1.3); without visual drive, neither stimulus would separate the sexes. Two stimuli thus decide between three ideas of what the muscle is for. **The animal can then correct the model:** in simulation, a search that reads only the noisy retinal movement improved the model's stimulus by 18–33 % within about 100 presentations, five minutes of experiment, for an "animal" whose love spot lies 20° higher than the model assumes.

Machines face the same conflict (Fig. 2c). In 150 simulated scenes, an event sensor registered a static scene in 0 % of cases without movement, in 89 % with a movement of half a pixel and in at least 98 % with two to six pixels. The same movement hid a small moving target: detection fell from 97 % at half a pixel to 44 % at one pixel and 7 % at two, and for a low-contrast target to 23 % already at half a pixel. No fixed movement serves both tasks. Either the movement follows the task, or self-generated events are told apart from the target's, which is what the efference copy does in the fly [14].

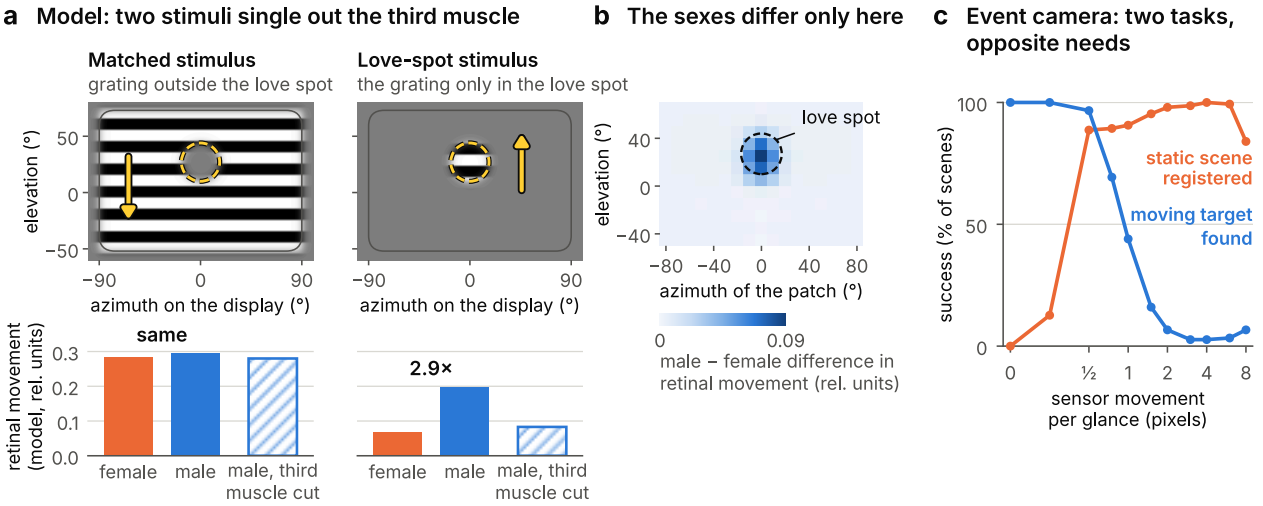


Figure 2. *Preliminary results.* (a) Two stimuli designed with the model for one bowl-shaped display (dashed circle: love spot; arrow: drift) and the predicted retinal movement of females, males and males whose third muscle is cut in the model. (b) Difference between the sexes for a small grating patch at each position of the display. (c) Event-camera simulation: share of 150 scenes in which a static scene is registered to a quarter of a pixel (orange) and a small moving target is found (blue), against the size of the sensor movement.

5. Objectives and hypotheses

O1. Model. Build a multi-tier, connectome-constrained spiking neural network (SNN) of the fly gaze loop, from photoreceptors to retinal displacement, *including electrical synapses*, in the female [12] and the male [39] *Drosophila* connectome. → **H1 (Sufficiency).** Chemical plus gap-junction connectivity from EM, with gap-junction weights set by apposition area, predicts motor-neuron tuning (± 15°), eye-movement direction and latency; a reduced SNN with ≥ 10× fewer state variables preserves these predictions.

O2. Test. Find out what the third muscle is for, in male and female *Eristalis*. → **H2 (A separate channel for the bright zone).** The third muscle is driven by motion inside the love spot, through the narrowed HSN, and moves the male retina in addition to, and independently of, the stabilising reflex that both sexes share. The experiments decide between this and two alternatives: an amplifier of the stabilising reflex, driven by wide-field motion, and a muscle without visual drive that moves the retina spontaneously. → **H2b (Origin).** The muscle is specified by its motor neuron, as the male-only Muscle of Lawrence of *Drosophila* forms only under *fruitless*-expressing motor neurons [42,43,79].

O3. Explain. → **H3 (Task optimality).** When connectome-constrained SNNs are trained on the male task (pursuit of small fast targets in clutter) or on the female task (hover stabilisation, feeding approach) under identical wiring priors and energy budgets [80], the male features (narrowed dorso-frontal integration, a third effector) emerge only under the pursuit objective, and the network uses the third effector as the male uses the muscle.

O4. Transfer. → **H4 (Transfer).** Event-based SNNs built from the extracted motifs, with task-dependent micro-movements of the sensor, outperform parameter- and energy-matched deep-network baselines on

small-target detection in clutter and on ego-motion estimation, and achieve closed-loop hovering and pursuit on an indoor quadrotor at a fraction of the energy.

The computer-vision question behind H4. Six of the seven predictions tested in WP4 (H4a–g in B2): (i) A sensor-movement policy tuned to the task beats a fixed one: the pursuit policy wins on small targets, the stabilising policy on ego-motion, and a fixed rotation [54] wins on neither. (ii) Feeding the motor command back into the motion units, the efference copy of [14], removes self-generated events better than software compensation. (iii) Linear electrical pooling cuts the latency from first event to motor command without loss of accuracy. (iv) Opponent coding with contrast normalisation makes velocity estimates robust to contrast and clutter [38]. (v) Under one energy budget, two task-specialised circuits with a shared front end beat one generalist. (vi) A sensor with a specialised zone gains from a second, independent movement of that zone: the machine version of the third muscle. Success is measured as accuracy per joule and as latency, against baselines matched in parameters and synaptic operations, under pre-registered protocols.

O5. Validate at scale and open everything. Every model runs on low-cost spiking robots in IU computer-vision courses and in a public experiment; students and citizens produce robustness data, proofread electrical synapses and collect hoverfly pursuit statistics.

6. Approach

WP1 Connectome → spiking circuit (M1–36). (a) MESH is scaled from 26 neurons to all wide-field motion neurons, retinal and neck motor neurons and optic-flow descending neurons, in the female and the male connectome; a 3D convolutional classifier trained on the validated sites ranks candidates, and community proofreading (WP5) confirms them. (b) Three model tiers: **Tier 1**, morphologically reduced biophysical neurons with electrical and chemical coupling, fitted by differentiable simulation [66]; **Tier 2**, integrate-and-fire neurons with gap-junction coupling, derived from Tier 1 by a reduction in which Kolmogorov–Arnold networks [93] return closed-form transfer functions; **Tier 3**, trainable SNNs masked by the connectome, trained with surrogate gradients [64,65]. (c) A motor readout maps muscle activation to retinal displacement [7,14]. Validation: motor-neuron imaging, eye tracking and the efference-copy recordings of [14].

WP2 What is the third muscle for? (M6–48). No hoverfly connectome exists, so male and female circuits are built by homology from the *Drosophila* model (the four HS cells, the VS cells and the motor neurons of the three muscles), constrained by our tomography atlas [2], dye fills and published hoverfly physiology [1,15,23]. This gives a family of male models that differ in what drives the third muscle and where it pulls. The experiments use one bowl-shaped projector display [22] and track the retinae of both eyes through the cornea, in three steps: (a) the matched and the love-spot stimulus in males and females; (b) a closed-loop search in which the measured retinal movement chooses the next stimulus, started from the model's proposal, so that the animal can correct the model; (c) electromyography of the three muscles, recordings and dye fills of the candidate cells, and section of the muscle as the causal test. A HOVER postdoc does this work in the Fenk laboratory (MPI for Biological Intelligence), which holds the only established retinal-tracking and retinal-muscle set-ups. *High risk, high gain:* retinal tracking during free-flight pursuit [21]. With R. Merrill (LMU), a HOVER PhD student traces the developmental origin of the muscle in pupae and its genetic basis (sex- and stage-resolved transcriptomics, the hoverfly genome [44]): H2b.

WP3 Theory: why a third muscle? (M12–54). In an embodied simulator with compound-eye optics and retinal actuators, Tier-3 networks are optimised for pursuit and for stabilisation under the same wiring prior and a spike-count energy penalty, both with the option of a third actuator [81,82]. We test whether the male features emerge and how the third actuator is used (H3), and extract the **minimal motif set** by ablation, mean-field analysis and Kolmogorov–Arnold readouts that turn the trained networks into equations.

WP4 Active event-based machine vision (M13–60). The motifs become event-driven SNNs on an active camera head: a wide-angle and a telephoto event camera, the machine counterparts of eye and bright zone, each on a piezo stage that moves its sensor by fractions of a pixel. Movement policies are fixed, fly-derived or learned, and an efference copy of each movement is fed to the network. Networks run on sensor-integrated and large-scale neuromorphic hardware [67,68,70]. Three pre-registered benchmarks: small-target detection and tracking in clutter against insect-inspired [61] and deep [94] detectors; ego-motion estimation on public event-camera datasets [58,59] against state-of-the-art artificial and spiking networks [55–57] and the fly-derived velocity CNN [38]; and closed-loop hovering and target pursuit on an indoor

event-camera quadrotor. Metrics: accuracy, latency, synaptic operations and measured energy. All applications are civil; code, data and the benchmark leaderboard are open.

WP5 Embodied validation at scale, teaching and open science (M1–60). All Tier-2 and Tier-3 models compile to low-cost two-wheeled camera robots that run spiking neurons natively. (a) Each semester, IU computer-vision students implement and test one HOVER motif; remote students receive a kit by mail. (b) A "robot zoo": schools, makerspaces and citizens run fly-type, female-type and male-type circuits on standard pursuit and stabilisation courses and upload logs: thousands of embodied trials under real-world optics and noise. (c) Community proofreading of candidate gap junctions in the MESH viewer [12,85]. (d) "Hoverfly watch": smartphone slow-motion videos of male pursuits supply the target statistics that define the WP3 tasks.

Feasibility. The pieces exist (section 4): MESH and its validated electrical synapses [14,41], a model from photoreceptor to eye muscle that already designs the WP2 stimuli, the closed-loop search and the event-camera simulation. The tomography atlas is in hand [2], hoverflies are reared year-round [19] and their genome is public [44]. I have shipped TinyML pipelines on classroom robots and teach computer vision to the cohorts that will run WP5. Hypothesis-critical experiments are done by HOVER team members hosted at the partner sites.

7. Why now, why me, why here

Why now. Whole-brain connectomes of both sexes, a validated pipeline for electrical synapses, differentiable biophysics, surrogate-gradient training, affordable event cameras, neuromorphic chips and classroom spiking robots have all matured within three years. Nobody has closed the loop.

Why me. My path runs through every level HOVER needs. I spent a decade recording from the fly motion-vision circuit, from the inputs of the elementary motion detector [95] to opponency and real-world velocity coding [26,38,96]. I then moved from one circuit to the whole wiring diagram, built MESH, performed the connectome analysis that revealed the electrical gaze pathway [14,41], and helped trace the evolution of insect eye movements and discover the third muscle [2]. In parallel I became a computer-vision scientist: segmentation and object recognition in terabyte electron-microscopy volumes, video-based event detection and TinyML classifiers on edge hardware, mobile eye-tracking with machine learning to read intended behaviour from human gaze [89] and, since 2026, Professor of Computer Vision at IU. I have also helped shape participatory research: first author of the Open Citizen Science framework [36], co-corresponding author of a plant electrophysiology study led by school pupils [88] and co-director of a research fellowship spanning more than 20 countries. HOVER is a new programme that I lead with my own team. It is not a continuation of my postdoctoral projects, nor of the Fenk laboratory's *Drosophila* programme on the mechanisms of eye movements, with which it is complementary: HOVER asks what a sex-specific eye muscle is for and what machines can learn from it, in a different species, with models and machines as the instruments.

Why here. IU is Germany's largest, largely distance-learning university. A computer-vision professorship there makes WP4 and WP5 core activities, with courses and a route to thousands of remote learners with kits at home. The partner laboratories supply what IU cannot: the only hoverfly retinal-muscle physiology set-ups and genomics for a non-model insect.

8. Expected outcomes and impact

(1) The first connectome-constrained spiking model of a complete visuomotor loop, with electrical synapses, from photon to eye movement, in both sexes of *Drosophila*. (2) The function, the neural drive and the developmental origin of the only known sexually dimorphic eye muscle, and with them a mechanistic and normative explanation of a visual sexual dimorphism. (3) A tested, open catalogue of fly-derived principles for active event-based vision, with a benchmark that lets the field measure "bio-inspired" claims. (4) An infrastructure through which students and citizens build and test spiking models. Beyond flies, HOVER is a template for asking what a circuit element is *for*: let models propose the decisive stimuli, let the animal correct them, and let a machine show whether the answer is useful.

Ambition and risk. HOVER joins three things that have not been joined before: a complete connectome-constrained motor loop with electrical synapses, a sex-specific effector as a natural experiment, and a machine-vision benchmark as the final arbiter. The riskiest elements, free-flight retinal tracking and the transfer hypothesis H4, are isolated so that their failure does not stop the rest; a rigorous negative result on H4 would still be a valuable, open contribution.

References (abbreviated; full list in `references.md`)

[1] Nordström *et al.* 2008 Curr Biol · [2] **Kviatko, …, Serbe-Kamp, …, Fenk** 2026 (manuscript) [verify status] · [7] Fenk *et al.* 2022 Nature · [12] Dorkenwald *et al.* 2024 Nature · [14] **Falt, Ammer, Serbe-Kamp *et al.*** 2026 bioRxiv · [15] Straw *et al.* 2006 J Exp Biol · [16] Nordström *et al.* 2006 PLoS Biol · [19] Nicholas *et al.* 2018 J Vis Exp · [21] Thyselius *et al.* 2023 J Exp Biol · [22] Prech *et al.* 2024 PLoS ONE · [23] Nicholas *et al.* 2026 eLife · [26] **Ammer, Serbe-Kamp *et al.*** 2023 Nat Neurosci · [30] Golan *et al.* 2020 PNAS · [36] **Serbe-Kamp *et al.*** 2023 RIO · [38] **Drews *et al.*** 2020 Curr Biol · [39] Berg *et al.* 2026 Cell · [41] **Serbe-Kamp** 2026 MESH software · [42] Lawrence & Johnston 1986 Cell · [43] Usui-Aoki *et al.* 2000 Nat Cell Biol · [44] *Eristalis tenax* genome, Darwin Tree of Life · [45] Lappalainen *et al.* 2024 Nature · [46] Shiu *et al.* 2024 Nature · [47] Schlegel *et al.* 2024 Nature · [48] Matsliah *et al.* 2024 Nature · [49] Bates *et al.* 2026 Nature · [50] Ammer *et al.* 2022 Curr Biol · [51] Juusola *et al.* 2017 eLife · [52] Kemppainen *et al.* 2022 PNAS · [53] Gallego *et al.* 2022 IEEE TPAMI · [54] He *et al.* 2024 Sci Robot · [55] Paredes-Vallés *et al.* 2024 Sci Robot · [56] Hagenaars *et al.* 2021 NeurIPS · [57] Gehrig *et al.* 2021 3DV · [58] Gehrig *et al.* 2021 IEEE RA-L · [59] Zhu *et al.* 2018 IEEE RA-L · [61] Wiederman *et al.* 2008 PLoS ONE · [62] Nordström & O'Carroll 2009 Trends Neurosci · [64] Neftci *et al.* 2019 IEEE SPM · [65] Zenke & Ganguli 2018 Neural Comput · [66] Deistler *et al.* 2025 Nat Methods · [67] Davies *et al.* 2021 Proc IEEE · [68] Mayr *et al.* 2019 arXiv · [70] Yao *et al.* 2024 Nat Commun · [71] Collett & Land 1975 · [72] Collett & Land 1978 · [74] Hausen & Strausfeld 1980 · [75] Gilbert & Strausfeld 1991 · [76] Hornstein *et al.* 2000 · [79] Gailey *et al.* 1991 Development · [80] Niven & Laughlin 2008 J Exp Biol · [81] Yamins & DiCarlo 2016 · [82] Lindsey *et al.* 2019 ICLR · [85] Kim *et al.* 2014 Nature · [88] **Madariaga, …, Serbe-Kamp, Marzullo** 2024 Plant Signal Behav · [89] **Serbe-Kamp *et al.*** 2026 PsyArXiv · [92] Hürkey *et al.* 2023 Nature · [93] Liu *et al.* 2024 (KAN) · [94] Gehrig & Scaramuzza 2023 CVPR · [95] **Serbe *et al.*** 2016 Neuron · [96] **Leonhardt, …, Serbe, …** 2016 Nat Neurosci · [108] Gehrig & Scaramuzza 2024 Nature · [109] Zhao *et al.* 2025 Nature · [110] Ponce *et al.* 2019 Cell · [111] Walker *et al.* 2019 Nat Neurosci · [112] Aloimonos *et al.* 1988 Int J Comput Vis

Curriculum Vitae and Track Record: Étienne Serbe-Kamp

Personal details

Prof. Dr. Étienne Serbe-Kamp · ORCID 0000-0002-7097-6274 · IU Internationale Hochschule, [campus / address], Germany · [IU e-mail] · github.com/ETigerschuss

Education and key qualifications

- **2016** PhD (Dr. rer. nat.), fast-track programme of the Graduate School of Systemic Neurosciences, LMU Munich, carried out at the Max Planck Institute of Neurobiology. Thesis: *[title]*. Defence: [DD.MM.2016] [verify, the date must match the certificate]. **Supervisor:** Prof. Alexander Borst.
- **2011** BSc Biology, LMU Munich.
- **Key qualifications.** In vivo whole-cell patch clamp, two-photon calcium imaging and optogenetics in the fly visual system; EM connectomics and research-software engineering (MESH); biophysical, reduced and differentiable circuit models; machine learning for stimulus design, segmentation and multimodal time series; edge AI and TinyML on robots; eye tracking and electrophysiology in humans and plants; design and evaluation of open citizen-science projects.

Current and previous positions

- **since 2026** Professor of Computer Vision (W2-equivalent, distance-learning programme Applied Artificial Intelligence), IU Internationale Hochschule, Germany. [start month; verify title]
- **2023–2026** Scientist, Active Sensing group (Dr. Lisa M. Fenk), Max Planck Institute for Biological Intelligence, Martinsried (12.8–50 %): connectome analysis, biophysical modelling, teaching of electrophysiology. [verify end date or ongoing guest status]
- **2026** Lecturer, AI and Bioinformatics (international master's programme, virtual and hybrid), SRH University Heidelberg (April–August).
- **2023–2024** Senior Scientist, Chair of Philosophy of Mind (Prof. Ophelia Deroy), LMU Munich (50 %): mobile eye tracking and machine learning on behavioural and survey data.
- **since 2020** Senior Scientist and Fellowship Co-Director, Backyard Brains Inc., Ann Arbor (USA) and Munich (30–75 %, industry; declared conflict of interest for WP5, see Part B2 and Resources). **2018–2019** R&D Scientist and Fellowship Mentor, Backyard Brains.
- **2016–2022** Postdoctoral researcher, Department Circuits–Computation–Models (Prof. Alexander Borst), Max Planck Institute of Neurobiology, Martinsried (full-time 2016; 25–50 % from 2018 in parallel with the industry position).
- **2011–2016** Doctoral researcher, same department.
- **PhD supervisor:** Alexander Borst. **Postdoctoral mentors:** Alexander Borst, Lisa M. Fenk.

Ten research outputs (most recent first)

Each entry states the significance, my role, and what it demonstrates for HOVER. Joint first authorship is marked *, joint corresponding authorship #.

1. *Falt T, Ammer G*, Serbe-Kamp É, Kroell LM, Friedrich AB, Guerreiro-Mota S, Šimsová E, Fenk LM (2026).** Vectorial efference copy and visuomotor transformation through gap junctions. *bioRxiv*, doi:10.64898/2026.08.11.744090 (posted 17 Aug 2026; in review [verify]). *Significance:* the motor neurons that move the fly retina are coupled to the HS and VS motion neurons by electrical synapses that current connectomes do not capture; the electrical wiring alone predicts motor tuning, the two-dimensional optokinetic reflex and an efference copy. *My role:* I built MESH and performed the connectome-scale contact-site analysis that defined the electrical wiring diagram (Figs. 3 and 4), and collected preliminary neurobiotin fills. *HOVER:* the basis of WP1 and of hypothesis H1.
2. **Serbe-Kamp É, Aliyeva A, Krick CB, Jantus CI, [complete author list exactly as posted] (2026).** Gaze on IPCC climate maps carries behaviour-specific signals for intended low-carbon action. *PsyArXiv*, osf.io/preprints/psyarxiv/64utk (posted Sep 2026; submitted to [journal]) [verify title, author order and DOI]. *Significance:* mobile eye-tracking during free viewing of climate maps, analysed with machine learning, predicts which low-carbon actions people intend. *My role:* first author; conceived the analysis, built the eye-tracking and classification pipeline, wrote the paper [verify]. *HOVER:* independent, first-author work on gaze and behaviour outside the fly field; the data-analysis pipeline carries over to retinal tracking in WP2.

3. **Serbe-Kamp É (2026).** MESH: connectome-scale contact-site detection and EM proofreading for putative gap junctions. Research software, MIT licence. github.com/ETigerschuss/MESH (development) and github.com/ActiveSensing/MESH (release v1.0.0 with a pre-built viewer). *Significance:* the first open pipeline that finds candidate electrical synapses in a whole-brain connectome, validates them in the EM volume and serves them for proofreading; 619 contact sites and 49 validated motor-neuron–LPTC sites for the 26-neuron gaze set, 2,895 EM images, public viewer. *My role:* sole developer. *HOVER:* WP1 and the community-proofreading strand of WP5.
4. **Kviatko P, Renous Y, Falt T, Kroell LM, Guerreiro Mota S, Pujari A, Rimniceanu M, Serbe-Kamp E, Šimsová E, Vollrath F, Wilde F, Hammel JU, Scheel M, Blanke A, Fenk LM (2026).** Spontaneous eye movements preceded visual stability reflexes in the evolution of insects. Manuscript [status: in review / to be posted on bioRxiv before 12 Jan 2027; verify; remove if not public]. *Significance:* comparative anatomy of 167 insect species and behaviour in 59 reconstruct the stepwise evolution of insect eye movements and reveal the male-only third retinal muscle of hoverflies. *My role:* [fill: micro-CT acquisition/segmentation, behavioural assays, analysis]. *HOVER:* the comparative basis of WP2 and hypothesis H2.
5. *Madariaga D, Arro D*, Irarrázaval C*, Soto A, Guerra F, Romero A, Ovalle F, Fedrigolli E, DesRosiers T, Serbe-Kamp É#, Marzullo TC# (2024).** A library of electrophysiological responses in plants: a model of transversal education and open science. *Plant Signal Behav* 19(1):2310977. doi:10.1080/15592324.2024.2310977 [verify the equal-contribution marks]. *Significance:* a library of electrical responses of more than 15 plant species to touch and heat, recorded by secondary-school and undergraduate students on open hardware; the pupils of a school in Santiago de Chile are first authors. *My role:* co-senior and co-corresponding author; designed the study and the recording protocol, supervised the students remotely, analysed the data and wrote the paper with T.M. [verify]. A second paper from this programme is in preparation [fill: status]. *HOVER:* the working model for WP5: participants as researchers and authors, not as an audience.
6. **Ammer G, Serbe-Kamp E, Mauss AS, Richter FG, Fendl S, Borst A (2023).** Multilevel visual motion opponency in *Drosophila*. *Nat Neurosci* 26:1894–1905. *Significance:* motion opponency is implemented at successive stages of the fly motion pathway, which sharpens flow-field selectivity. *My role:* second author; [fill: two-photon imaging and electrophysiology of T4/T5 and LPTCs, analysis, figures; mark joint first authorship if applicable]. *HOVER:* opponency is one of the candidate motifs transferred in WP4.
7. **Serbe-Kamp É, Bemme J, Pollak D, Mayer K (2023).** Open Citizen Science: fostering open knowledge with participation. *Res Ideas Outcomes* 9:e96476. doi:10.3897/rio.9.e96476. *Significance:* defines Open Citizen Science as a methodological basis that joins open-science practice with participation, with two Wikimedia Open Science Fellowship projects as case studies, including ERGo!, my low-cost electroretinogram platform. *My role:* first and corresponding author. *HOVER:* the framework of WP5.
8. **Drews M, Leonhardt A, Pirogova N, Richter F, Schuetzenberger A, Serbe E, Braun L, Borst A (2020).** Dynamic signal compression for robust motion vision in flies. *Curr Biol* 30:209–221 [verify author order]. *Significance:* a fly-like velocity-estimation network trained on natural scenes (`deepfly`) shows that contrast normalisation makes motion vision robust. *My role:* [fill]. *HOVER:* the CNN baseline in WP4 and a motif candidate.
9. **Serbe E, Meier M, Leonhardt A, Borst A (2016).** Comprehensive characterization of the major presynaptic elements to the *Drosophila* OFF motion detector. *Neuron* 89:829–841. *Significance:* the first complete functional characterisation of the inputs to the OFF elementary motion detector by in vivo patch clamp. *My role:* first author; all recordings, analysis and writing. *HOVER:* the physiological ground truth for the Tier-1 front end.
10. **Leonhardt A, Ammer G, Meier M, Serbe E, Bahl A, Borst A (2016).** Asymmetry of *Drosophila* ON and OFF motion detectors enhances real-world velocity estimation. *Nat Neurosci* 19:706–715. *Significance:* a biological design choice, the ON/OFF asymmetry, improves velocity estimation in natural scenes, which is exactly the logic of HOVER's H4. *My role:* [fill: electrophysiology].

Further outputs: co-author of the directional tuning map of elementary motion detectors (Maisak *et al.* 2013 *Nature*), of contrast vision (Bahl *et al.* 2015 *Neuron*), feedforward inhibition (Mauss *et al.* 2014 *J Neurosci*) and complementary mechanisms of direction selectivity (Haag *et al.* 2016 *eLife*); more than 20 peer-reviewed papers in total.

Selected examples of significant peer recognition

- **2026** Appointment as Professor of Computer Vision, IU Internationale Hochschule, after a competitive call (Applied Artificial Intelligence programme).
- **2026** Keynote lecture, GSN Student Conference, Graduate School of Systemic Neurosciences, Munich [verify].
- **2025** Invited speaker, international conference *Citizen Science: Bridging Theory and Practice*, Vilnius University (talk: *Citizen Science, in and out of the Lab*); panellist, European Citizen Science Association webinar *AI and Citizen Science*.
- **2024** LMUexcellence Innovation Fund, LMU Munich [role and amount; verify].
- **2022** Member of the science jury of *IdeenLauf*, the citizen-participation initiative of the German Federal Ministry of Education and Research (BMBF) in the Science Year 2022.
- **2022–2023** Principal investigator of two EU Horizon Europe proposals scored 14/15 and 14.5/15 [state whether funded; verify].
- **2020** Chen Innovator Grant, Tianqiao and Chrissy Chen Institute, Caltech [role; verify].
- **2024–** Scientific lead for modelling and machine learning on the NIH SBIR phase II project *SpikerBots* (brain-based neurorobots for schools; PI G. Gage, Backyard Brains) [verify].
- **[year]** Fellow, Open Science Fellows Program (*Fellow-Programm Freies Wissen*), Wikimedia Deutschland and Stifterverband [verify year]. **[year]** Young Scientist Award [fill body].
- Media: scientist in season 2 of the ZDF/KiKA series *Princess of Science* (the series received the Grimme-Preis), ProSieben *Galileo*, Radio Television of Serbia; more than 50 public science events on four continents.

Supervision, teaching and mentoring

- Co-director (since 2020; mentor since 2018) of the Backyard Brains Fellowship, an international research programme: more than 30 fellows from more than 20 countries, supervised almost entirely online; fellowship projects have led to peer-reviewed papers with fellows as authors [n; verify].
- Supervision of [n] BSc and MSc theses and co-supervision of [n] doctoral researchers [fill].
- Teaching: computer-vision modules at IU (since 2026); applied AI, computer vision and statistics in the international master's programme at SRH Heidelberg (2026); lecturer at the Munich Neurocamp (2019, 2025); summer schools and workshops since 2011, including more than 50 school and public workshops on four continents.

Invited talks and lectures (selection)

2026 keynote, GSN Student Conference, Munich [verify] · 2025 international citizen-science conference, Vilnius University · 2025 ECSA webinar on AI and citizen science · 2024 VISTA Science Academy, Klosterneuburg · 2023 *Conduction Unveiled*, international online lecture · 2022 SENSE festival, BIOTOPIA Munich · 2021 Wikimedia Winter School · 2020 innOsci *Innovations in Open Science*, Berlin · 2019 Hirnkastl workshop, Cambridge · 2017 Brain Awareness Week, Montevideo.

Funding obtained or co-obtained

LMUexcellence Innovation Fund (2024) [role, amount] · Chen Innovator Grant, Caltech (2020) [role] · Open Science Fellowship, Wikimedia Deutschland [year, amount] · scientific lead for modelling on the NIH SBIR phase II project *SpikerBots* (PI G. Gage) [verify] · PI of two EU Horizon Europe proposals (2022, 2023) scored 14/15 and 14.5/15 [funded? verify].

Career path and life events

My career combines academic research (Max Planck Society, LMU), industry (Backyard Brains, a neuroscience-education company, where I co-direct an international research fellowship) and science communication. From 2018 I held the Max Planck position part-time alongside the industry role, which is why the postdoctoral period spans six calendar years. This combination is the capacity that WP4 and WP5 require: I have shipped edge-AI products, mentored more than 30 scientists and engineers in more than 20 countries almost entirely online, and turned public participation into peer-reviewed research. [Add parental leave or other life events if applicable.]

Noteworthy contributions to the research community

- **Open research tools:** MESH (connectome contact detection and EM proofreading, with pre-built releases that need no installation); ERGo! (open electroretinogram platform); open models of the fly motion pathway. Co-development of the NIH-funded Computational SpikerBox and SpikerBot, low-cost platforms on which pupils and students run spiking neurons and computer vision.
- **Participatory research:** co-founder of the Hirnkastl Science Collective; author of the Open Citizen Science framework; a published study with school pupils as first authors; Open Science ambassador (BMBF IdeenLauf). Lecturer at the Munich Neurocamp (2019, 2025) and in international summer schools since 2017. [Add reviewing and editorial activities.]

Research independence and the capacity to deliver HOVER

My work spans the levels HOVER connects: cellular physiology of fly motion vision (first-author *Neuron* 2016; *Nature Neuroscience* 2016, 2023), connectomics and research software (MESH, the electrical gaze pathway), evolution (the origin of insect eye movements and the male-only third retinal muscle), differentiable modelling and machine learning (stimulus optimisation, segmentation of terabyte EM volumes, TinyML on robots), human gaze (first-author eye-tracking work at LMU) and participatory science (first and senior-author publications, a national jury, international invited talks). The independent lines are visible: MESH is my own tool, the citizen-science programme is my own, the eye-tracking study is my own, and the gap-junction search strategy that made the electrical gaze pathway findable was my design. HOVER is a new programme defined and led from my group at IU. It moves from describing circuits to asking *why* circuits differ between the sexes, in a new species, with connectome-constrained spiking models, neuromorphic machines and public experiments as the instruments. It is complementary to, and agreed with, the Fenk laboratory's *Drosophila* programme on eye-movement mechanisms. The partner groups contribute set-ups and expertise; the team members at the partner sites are selected, funded and directed by me.